



Figure 2: msgbox widget with title texts



Figure 3: Pause widget with --color option

- **--colors** – With this, we can change the video attributes of the Dialog titles' text.
- **--insecure** – By default, the password widget created by Dialog doesn't echo what we type on the text console screen. We can see typed characters echoed as asterisks, using this option.

Let's now start playing with the Dialog widgets. The latest version provides the following widgets: calendar, checklist, dselect, editbox, form, fselect, gauge, infobox, inputbox, menu, mixedform, mixedgauge, msgbox, passwordbox, passwordform, pause, progressbox, radiolist, tailbox, tailboxbg, textbox, timebox, yesno, etc.

Let's explore some of these now. Run the following command on a console:

```
dialog --infobox "This is just an infobox widget from dialog." 0 0
```

As you can see in Figure 1, this will draw an infobox widget with the information we specified.

After creating the infobox widget, Dialog immediately returns the console prompt. As you probably have guessed, this widget's purpose is to display information and quit immediately. I've passed two zeros in the above command for the height and width of the widget—passing 0 makes Dialog take care of the widget's height and width. You can consider 0 0 for the height and width, respectively, as very

safe values, most of the times.

You'll also notice in Figure 1 that the infobox widget doesn't have any title text, neither on the message box nor on the background. Let's add the title in our next example:

```
dialog --backtitle "Message Box" --title msgbox \
--msgbox "This is a msgbox widget from dialog." 0 0
```

Can you see the title texts in Figure 2 now? Also, if you press **Esc**, **Enter** or left-click on the OK button, then Dialog returns. We will look into these actions when we discuss the uses of Dialog widgets in shell scripts, later in the article.

Till now we have only looked at some boring default colours for texts. We can change the attributes and colours of these text messages passed to Dialog simply by embedding **'Z'** sequences in different texts passed to Dialog, along with the **--colors** option. Enter the following command to see it all in action:

```
dialog --colors --backtitle "\Zr\ZSpause\Zr" --title "\Zr\ZSpause\Zr" \
--pause "\Zb\ZGThis is a pause widget from dialog." 10 28 15
```

Can you see some colourful texts in and around the pause widget as shown in Figure 3? A pause widget shows a meter bar that keeps on decreasing with every elapsed second. If you don't press **Esc**, or hit the OK/Cancel buttons during the seconds remaining, the widget returns once the timeout value passed to it has elapsed. The various text attributes and colour values with embedded **'Z'** sequences in various title strings are:

- **r**: reverse
- **b**: bold
- **u**: underline
- **n**: restore normal settings
- **0**: black
- **1**: red
- **2**: green
- **3**: yellow
- **4**: blue
- **5**: magenta
- **6**: cyan
- **7**: white

We can also change the default titles of the **OK** and **Cancel** buttons, or show an extra button with the desired text in the widgets. Enter the following command to see these processes in action:

```
dialog --backtitle Calendar --title calendar --extra-button \
--extra-label "Extra step" --ok-label "Ok computer" \
--exit-label "bye bye" --calendar "Please select a date" 0 0
```

Take a look at Figure 4. We have inserted a new button and customised the texts of all the buttons here. We can move between different sub windows of the calendar widget using **Tab**. If you select to click on any of the buttons except **Esc** or hit the **Cancel** button, then you will see the selected date printed on the widget itself. This is the default